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MGR INŻ. OLEKSANDR YEROSHKIN, 24.10.2023

Mobile Applications for IOS

Apple Human Interface Guideline by examples

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PROGRAMME CONTENT		
Lecture		Number of hours
Lec 1	Introduction. Review of selected mobile applications. Introduction to Apple devices and Mac OS.	2
Lec 2-3	UI and UX. Interactions. Touch screens. Material design. Human Interface guidelines. Voice communication. Siri.	4
Lec 4	Tools for multimedia processing. Gathering multimedia data. Copyright.	2
Lec 5	Implementation rules in Xcode. The structure of application written in Swift.	2
Lec 6-7	Presentation of basic elements of Swift by examples. Core libraries.	4
Lec 8	Apple Human Interface Guideline by examples.	2
Lec 9	Libraries and frameworks supporting creation of multimedia applications. Short characteristics of Kotlin language.	2
Lec 10	Applications of augmented reality. Code analysis of application with augmented reality mechanisms.	2
Lec 11	Games review. Connections between games and the progress of algorithms and programming languages.	2
Lec 12	Multimedia data compression. Compression formats. Video and audio streaming.	2
Lec 13	Multimedia in mobile systems. Cameras. Recommended frameworks.	2
Lec 14	Interaction mechanisms – review. New technologies and multimedia devices.	2
Lec 15	Summary. Perspectives of multimedia techniques.	2
	Total hours	30



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Filter

- > Getting started
- > Foundations
- > Patterns
- > Components
- > Inputs
- > Technologies

Human Interface Guidelines

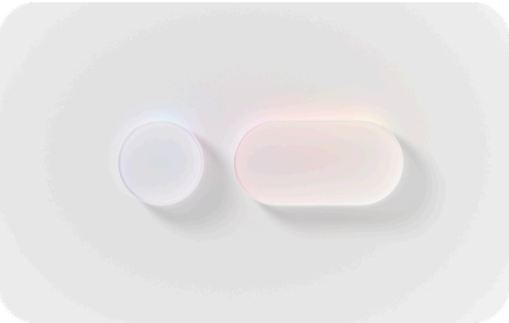
The HIG contains guidance and best practices that can help you design a great experience for any Apple platform.

As you design interfaces for Apple platforms, keep these principles in mind:



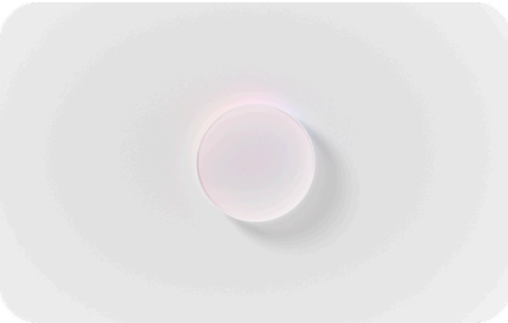
Hierarchy

Establish a clear visual hierarchy where controls and interface elements elevate and distinguish the content beneath them.



Harmony

Align with the concentric design of the hardware and software to create harmony between interface elements, system experiences, and devices.



Consistency

Adopt platform conventions to maintain a consistent design that continuously adapts across window sizes and displays.

For developer guidance, see [Adopting Liquid Glass](#).

Design fundamentals





Developer

News

Discover

Design

Develop

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Design

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Filter

Designing for tvOS

Designing for visionOS

Designing for watchOS

> Foundations

> Patterns

> Components

> Content

> Layout and organization

> Menus and actions

> Navigation and search

> Presentation

> Selection and input

> Status

> System experiences

App Shortcuts

Complications

Home Screen quick actions

Live Activities

The menu bar

Notifications

Status bars

Top Shelf

Watch faces

Widgets

Components

Learn how to use and customize system-defined components to give people a familiar and consistent experience.

Content

Layout and organization

Menus and actions

Navigation and search

Presentation

Selection and input

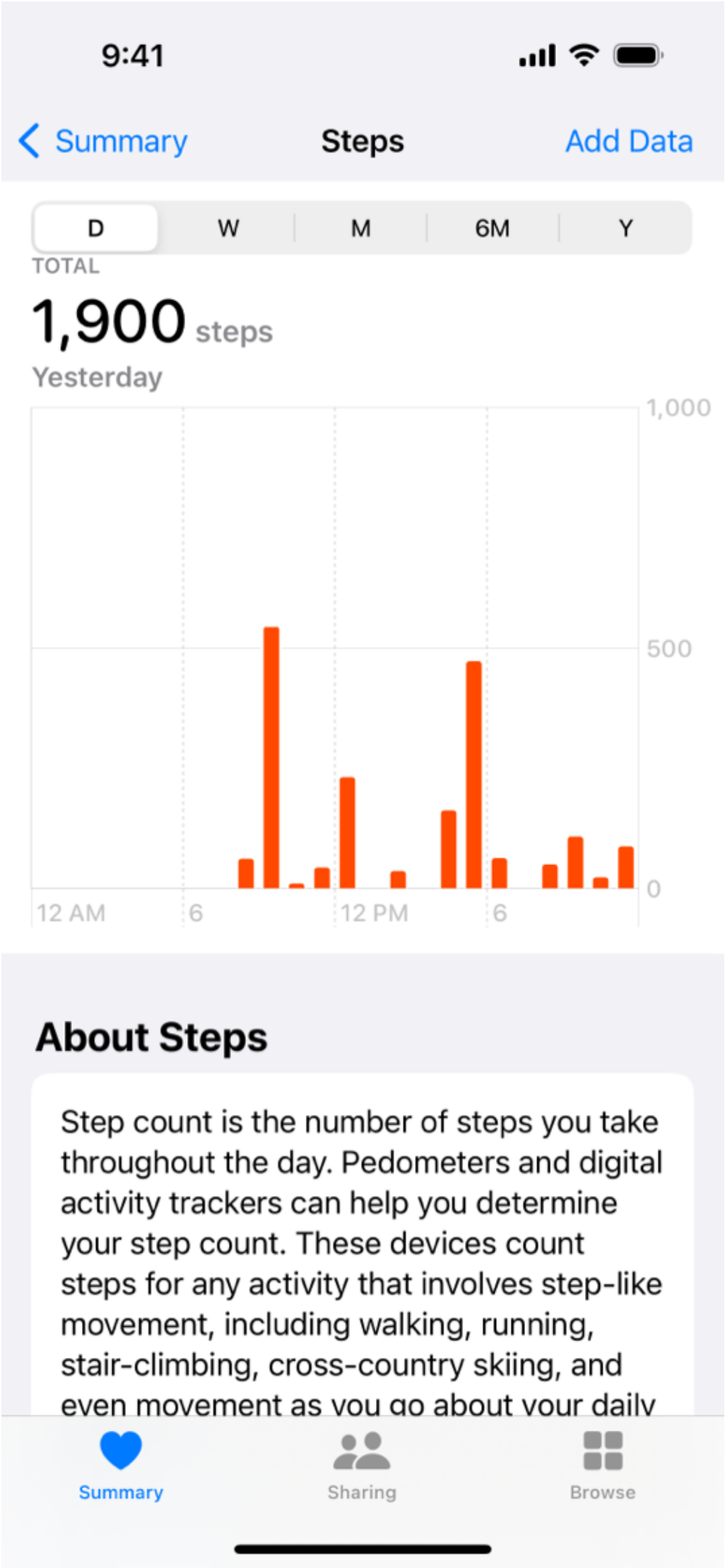
Status

System experiences

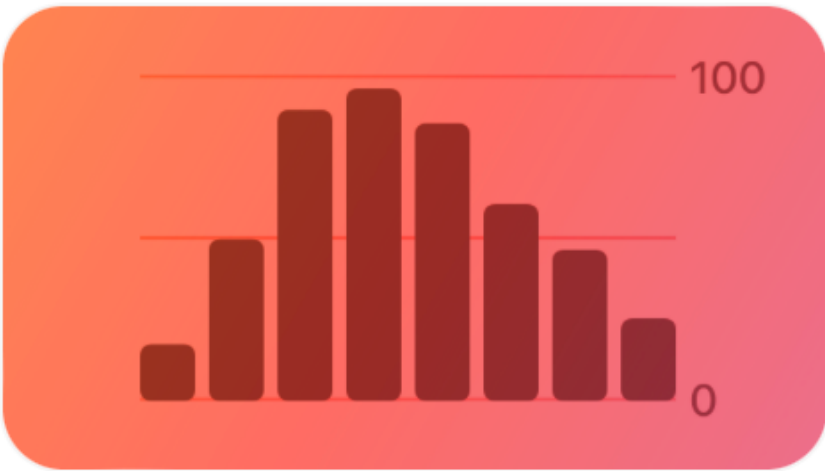
<https://developer.apple.com/design/human-interface-guidelines/components>



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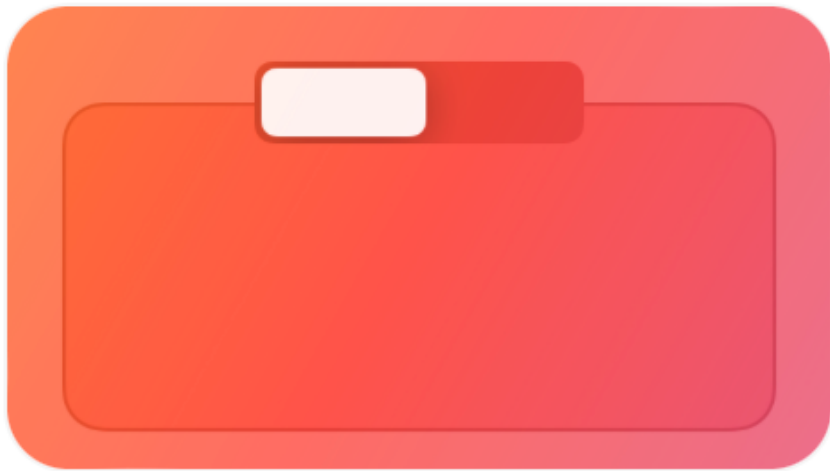
Apple Health



Charts

The interface is not merely a visual display—in fact, it's possible to have an interface with no visual display at all. A human interface is the sum of all communication between the computer and user.

Text views



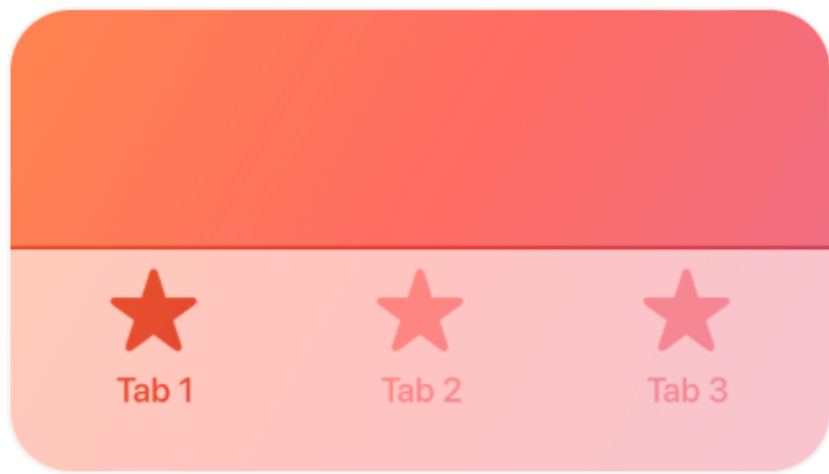
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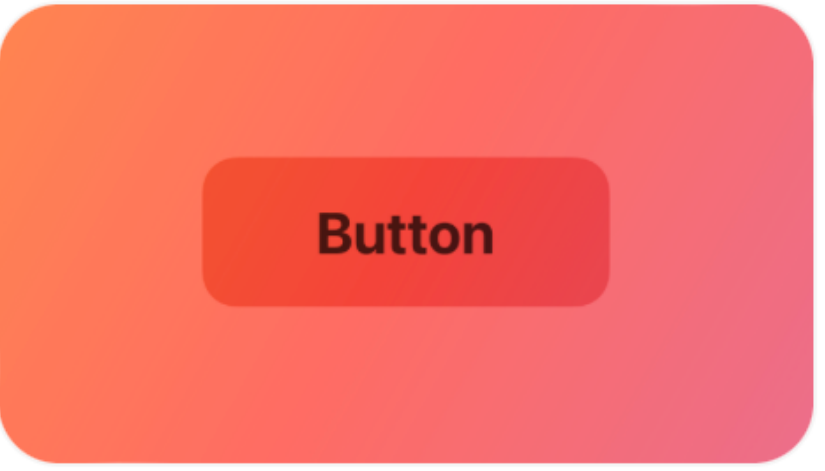
Labels

< Parent Title
Large Title

Navigation bars



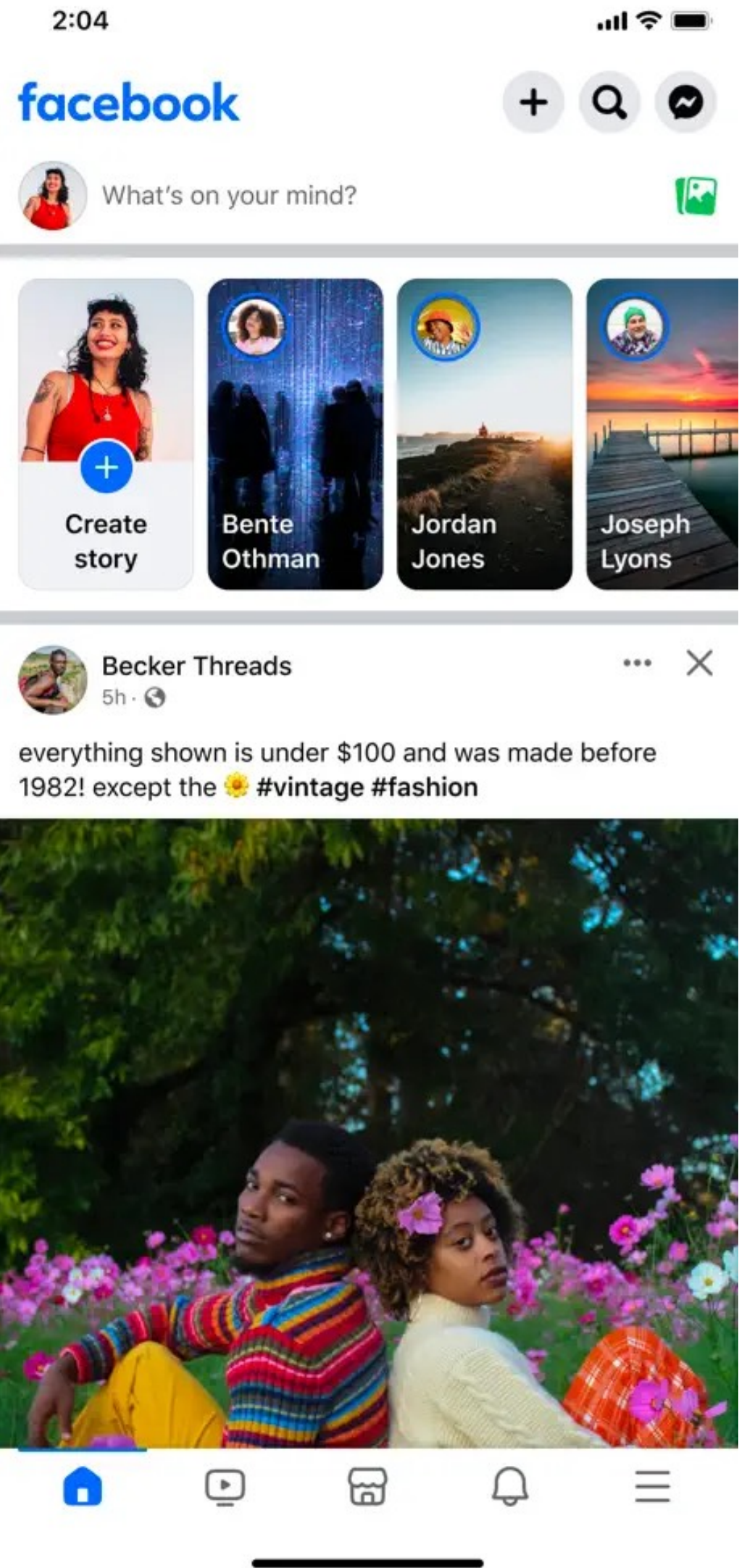
Tab bars



Buttons

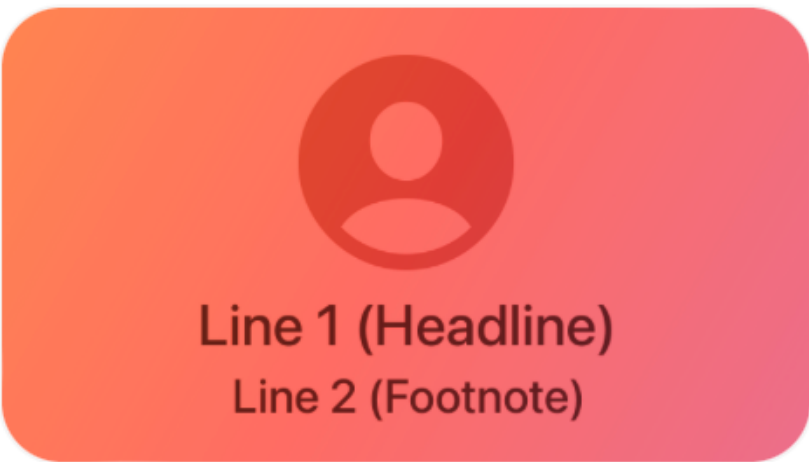


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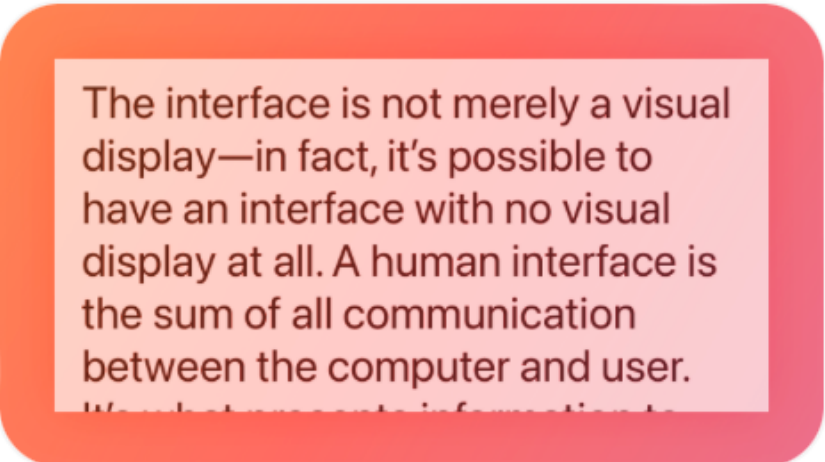


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Facebook



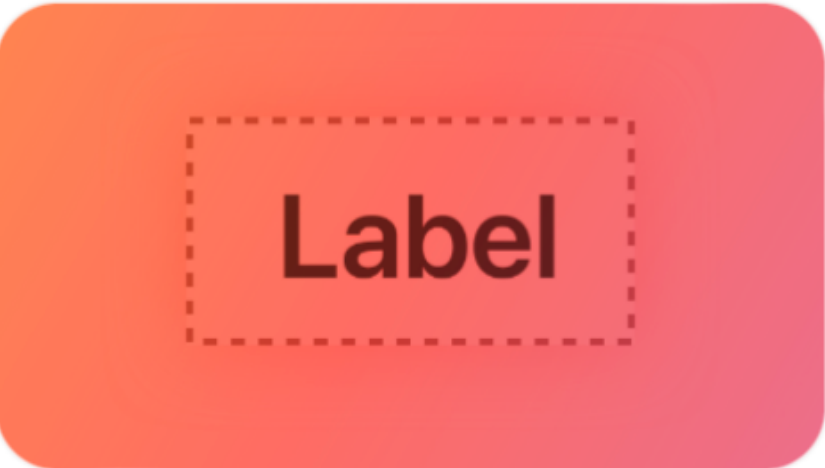
Lockups



Text views



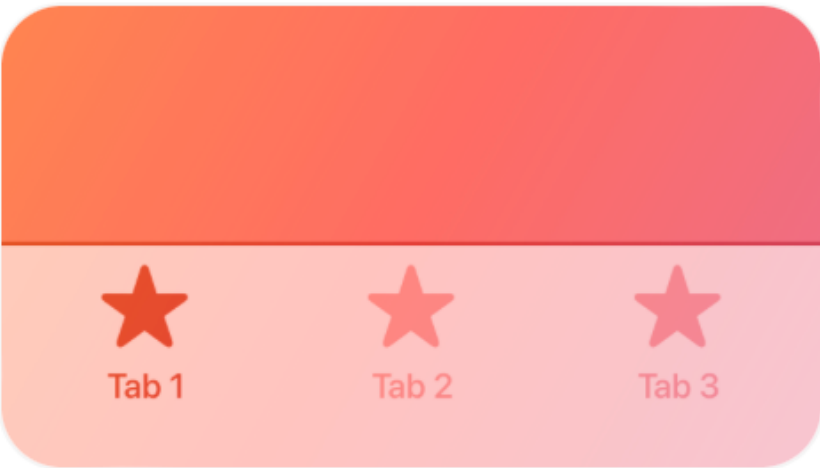
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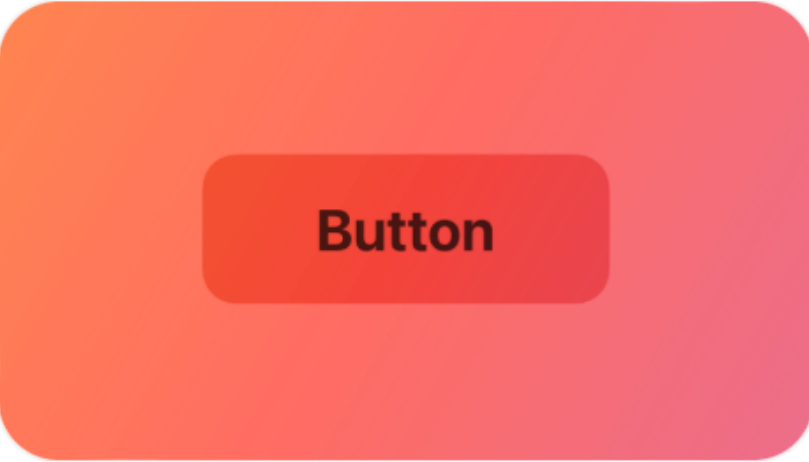
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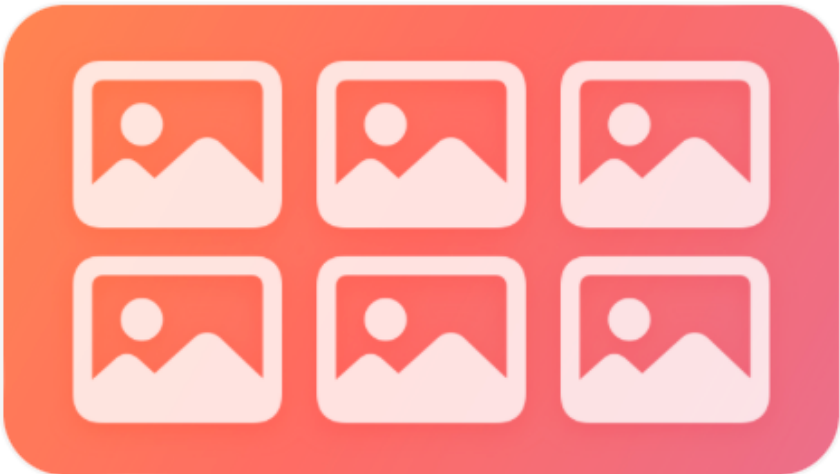
Navigation bars



Tab bars



Buttons



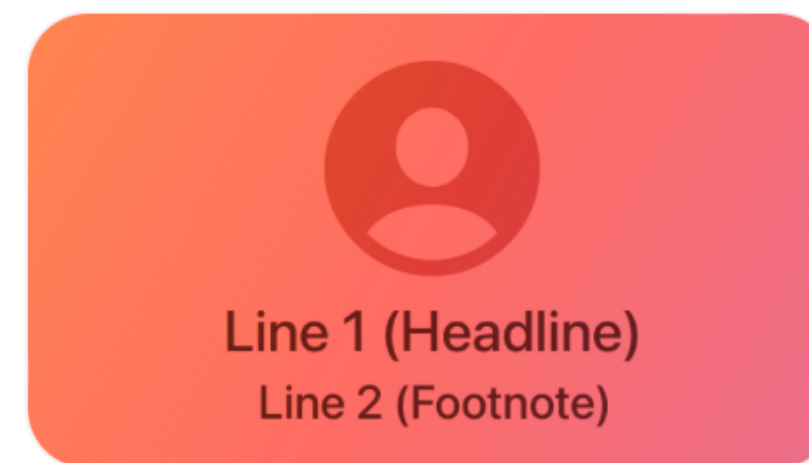
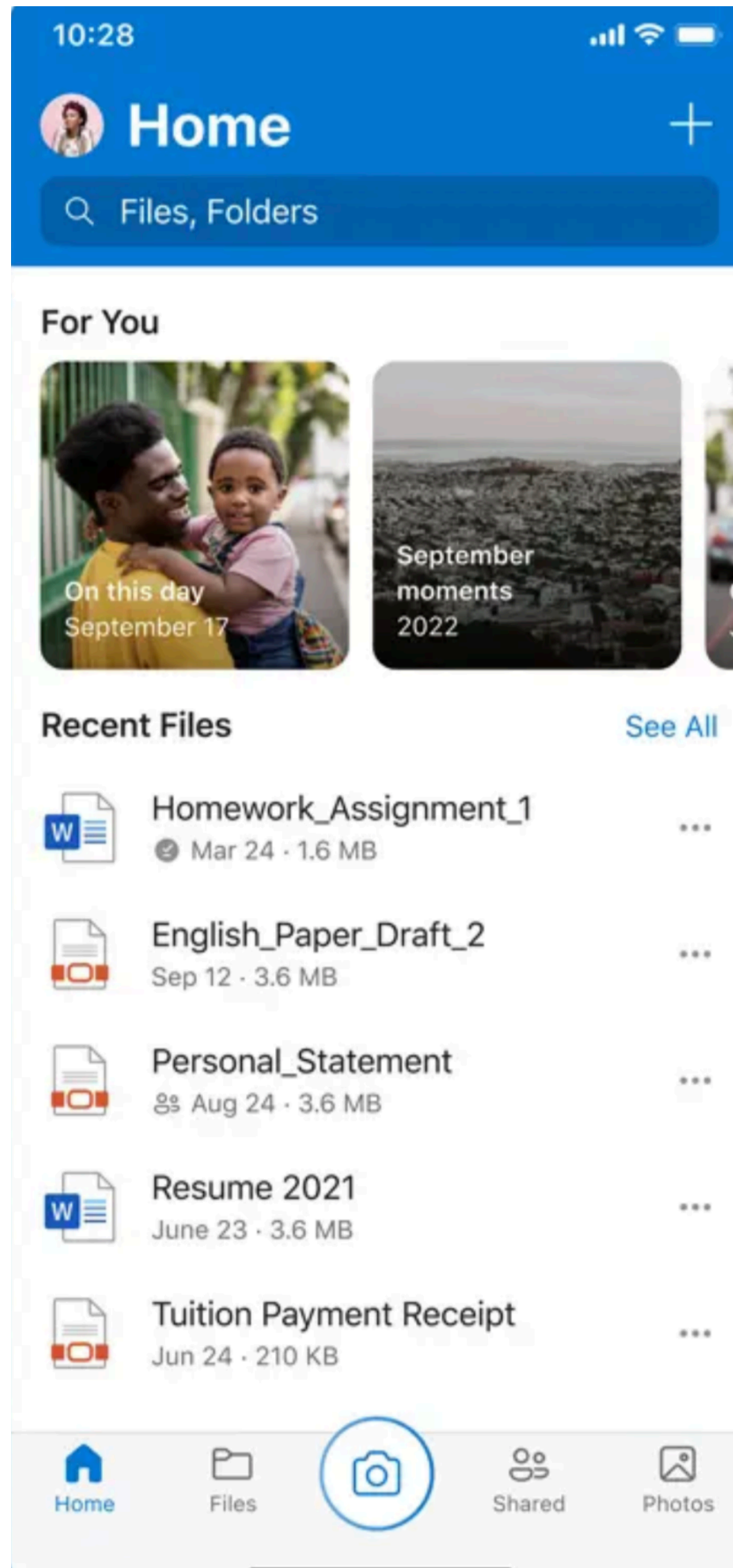
Collections



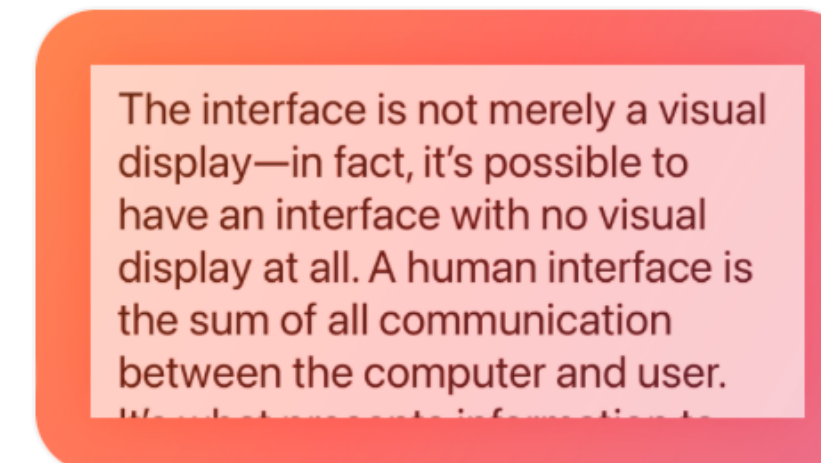
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Microsoft OneDrive



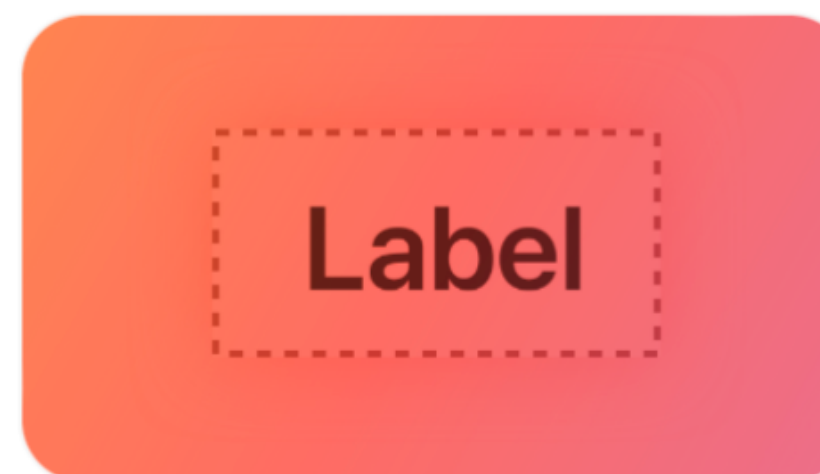
Lockups



Text views



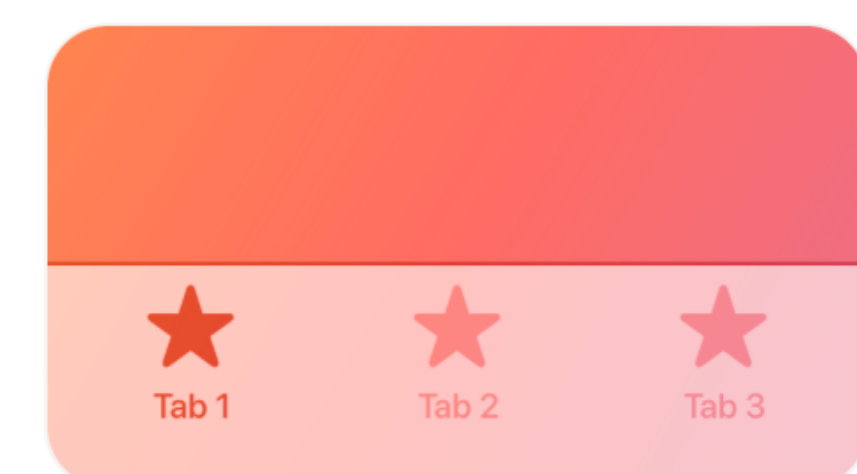
Image views



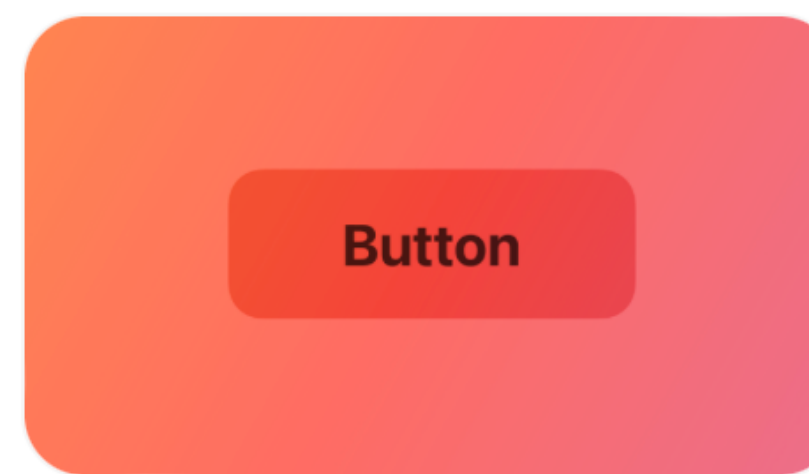
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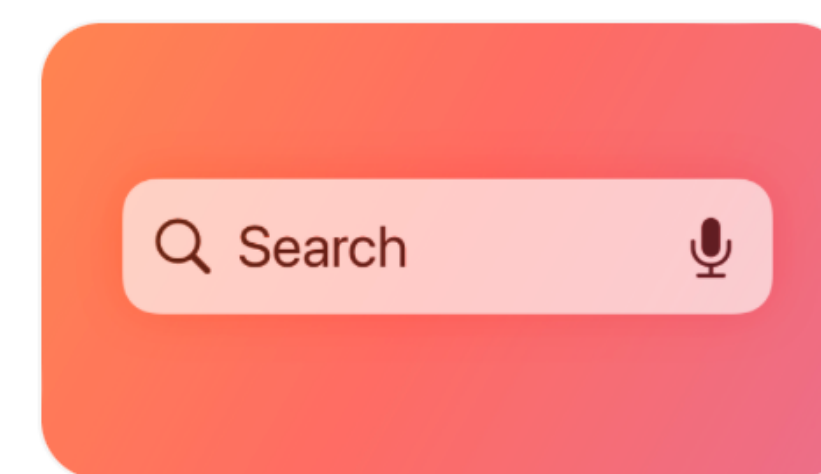
Navigation bars



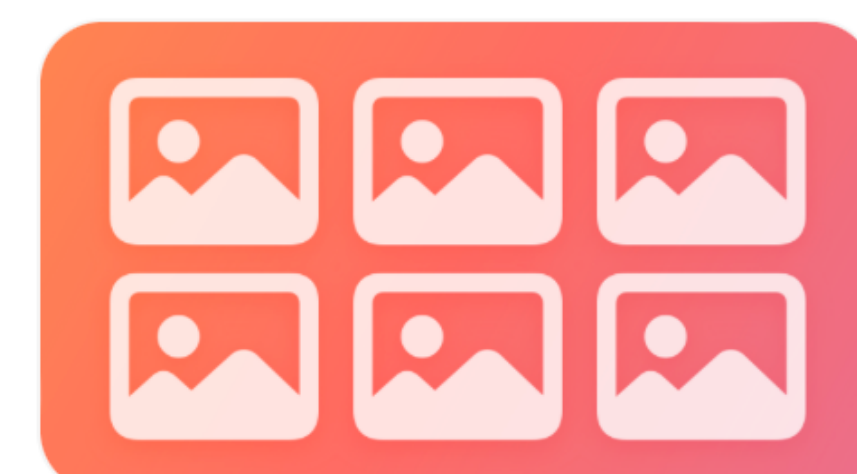
Tab bars



Buttons



Search fields

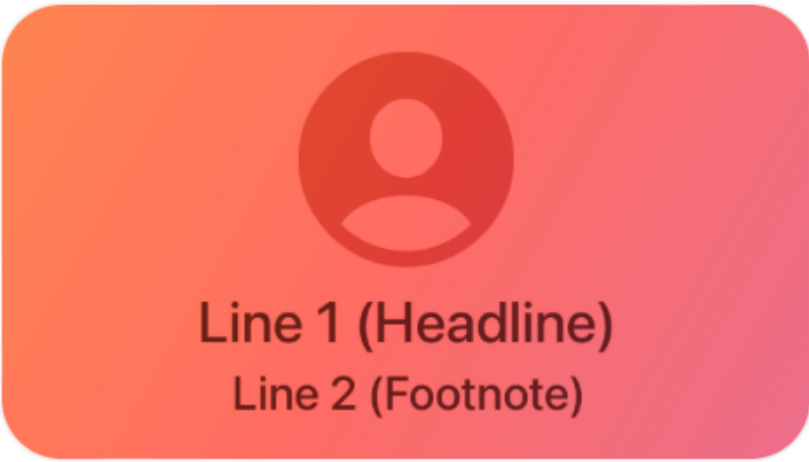
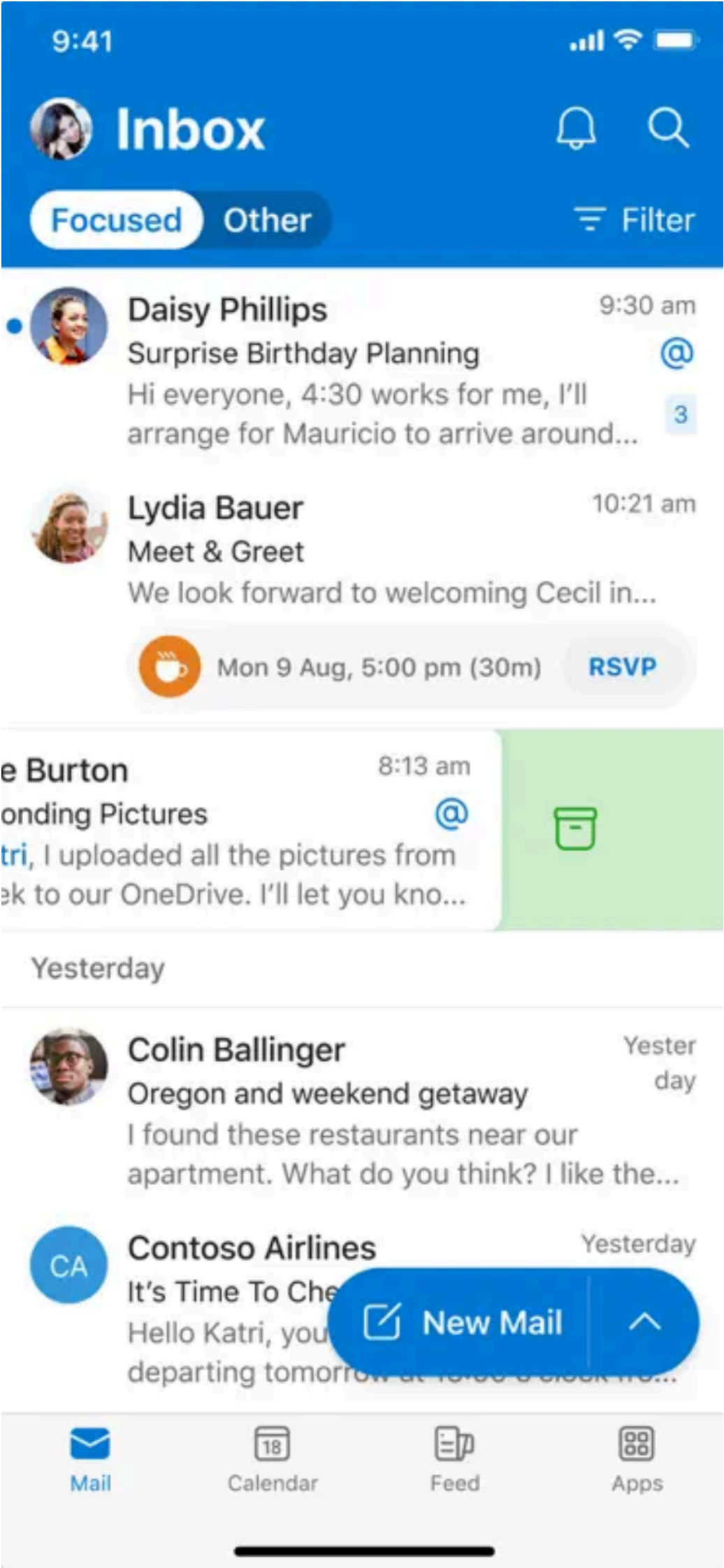


Collections

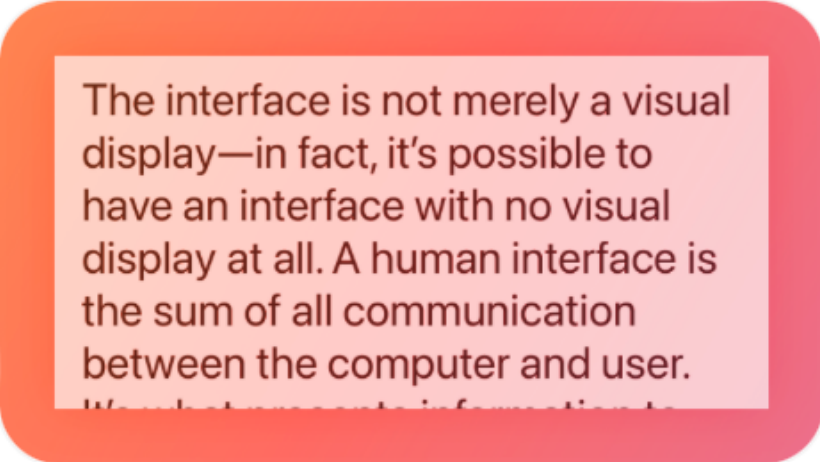


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Microsoft Outlook



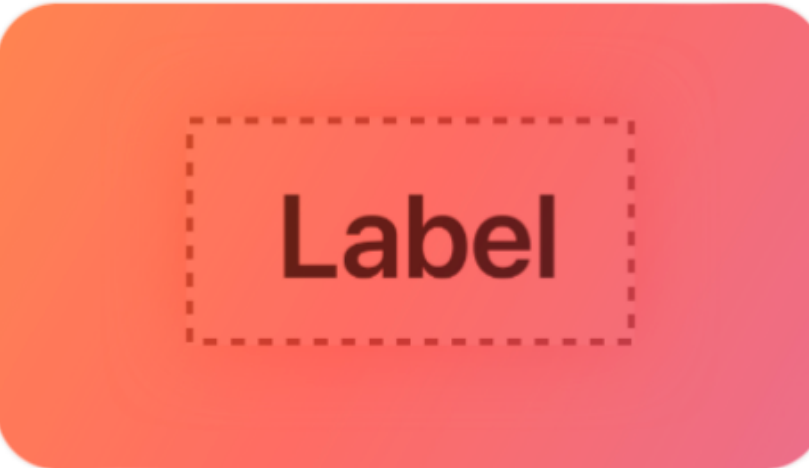
Lockups



Text views



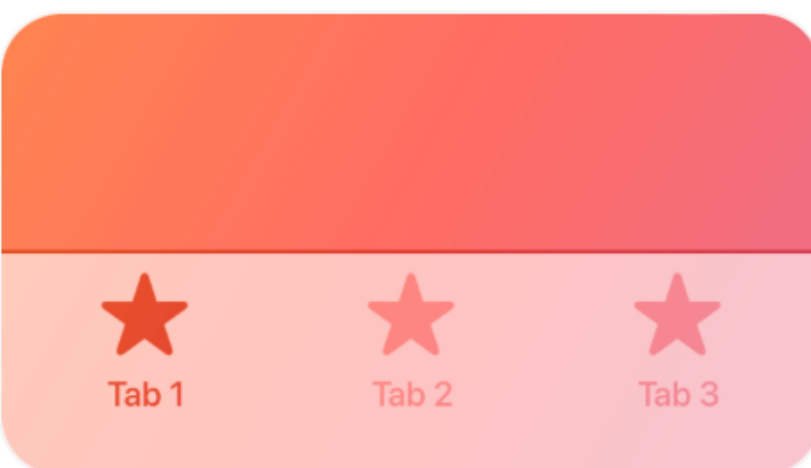
Image views



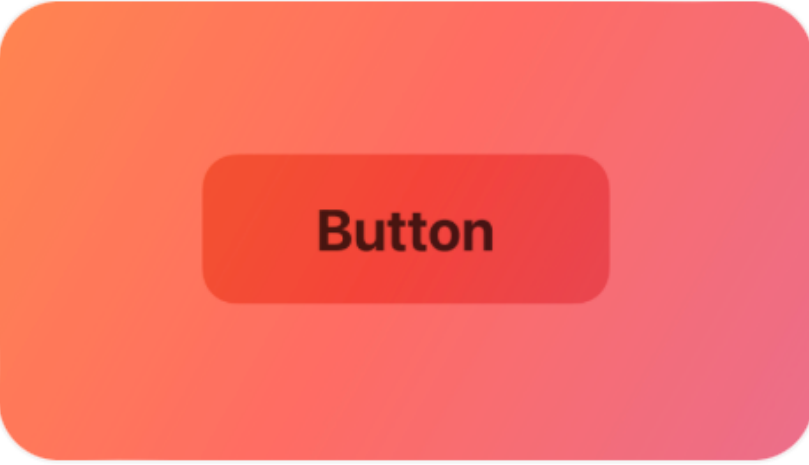
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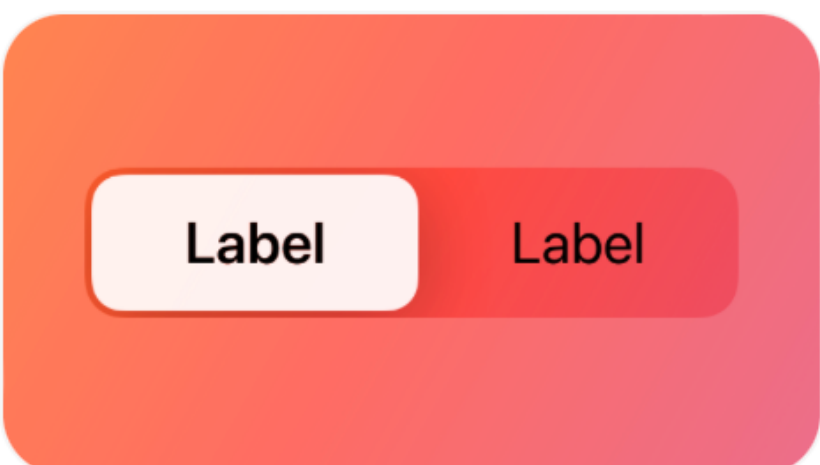
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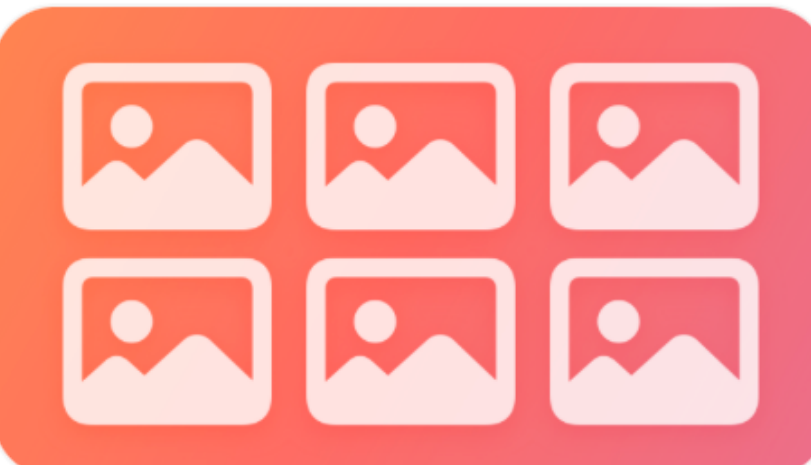
Tab bars



Buttons



Segmented controls



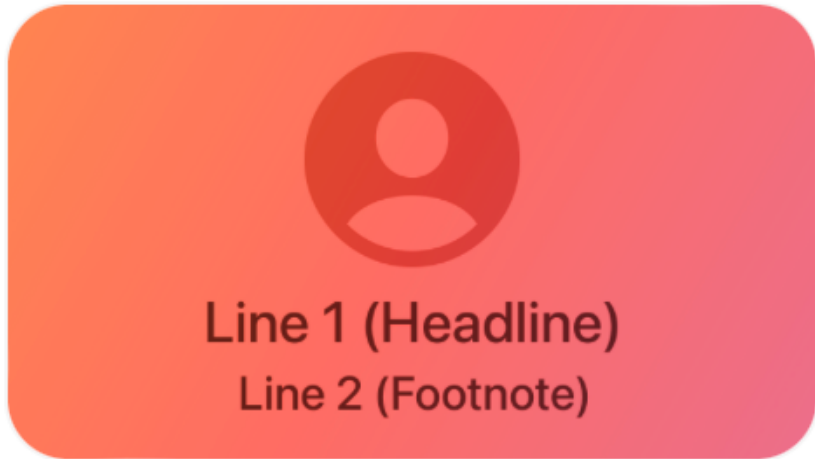
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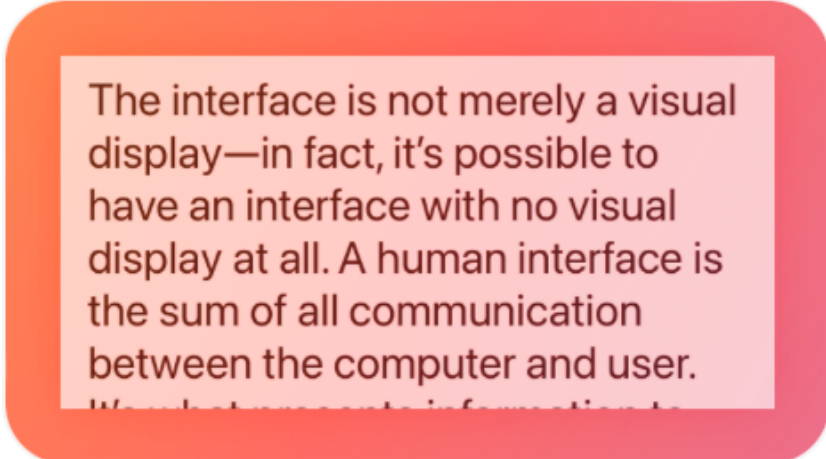
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Spotify



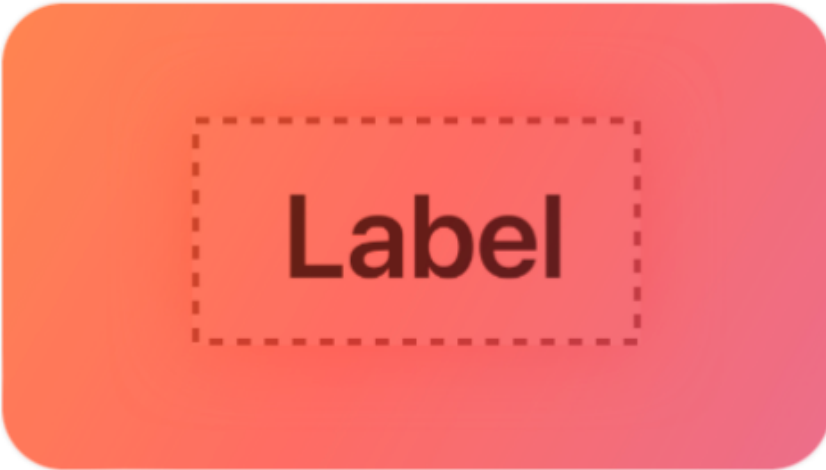
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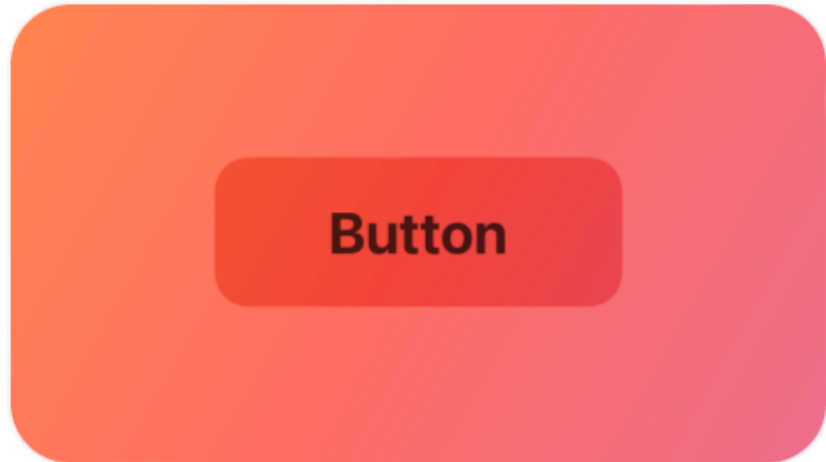
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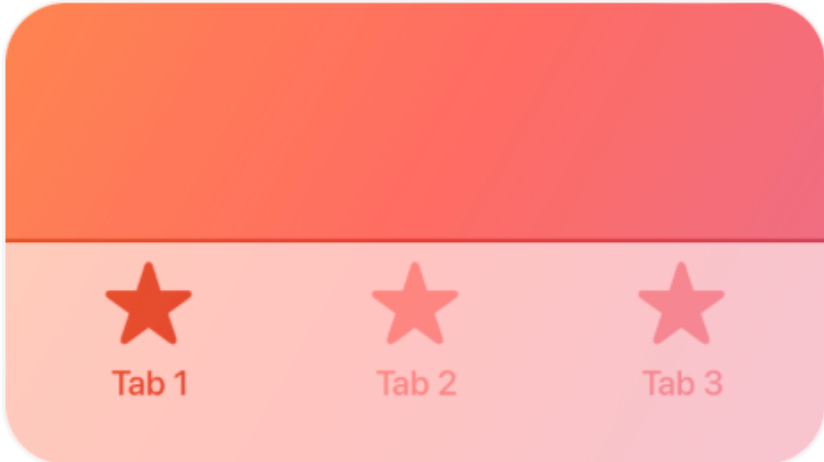
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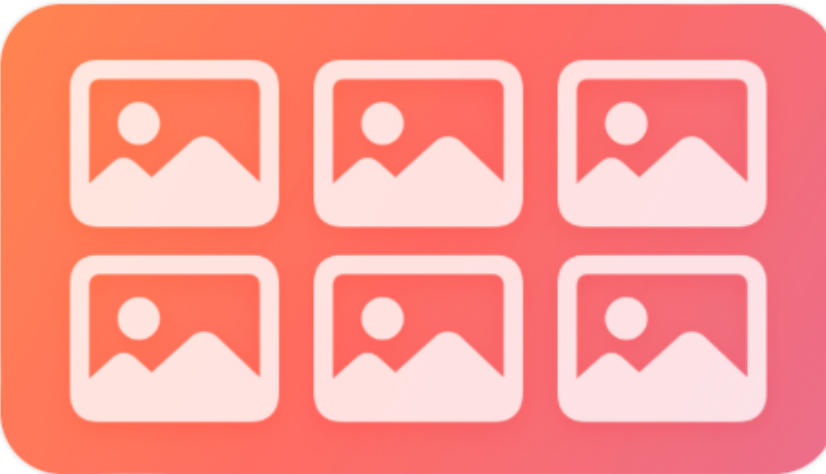
Labels



Buttons



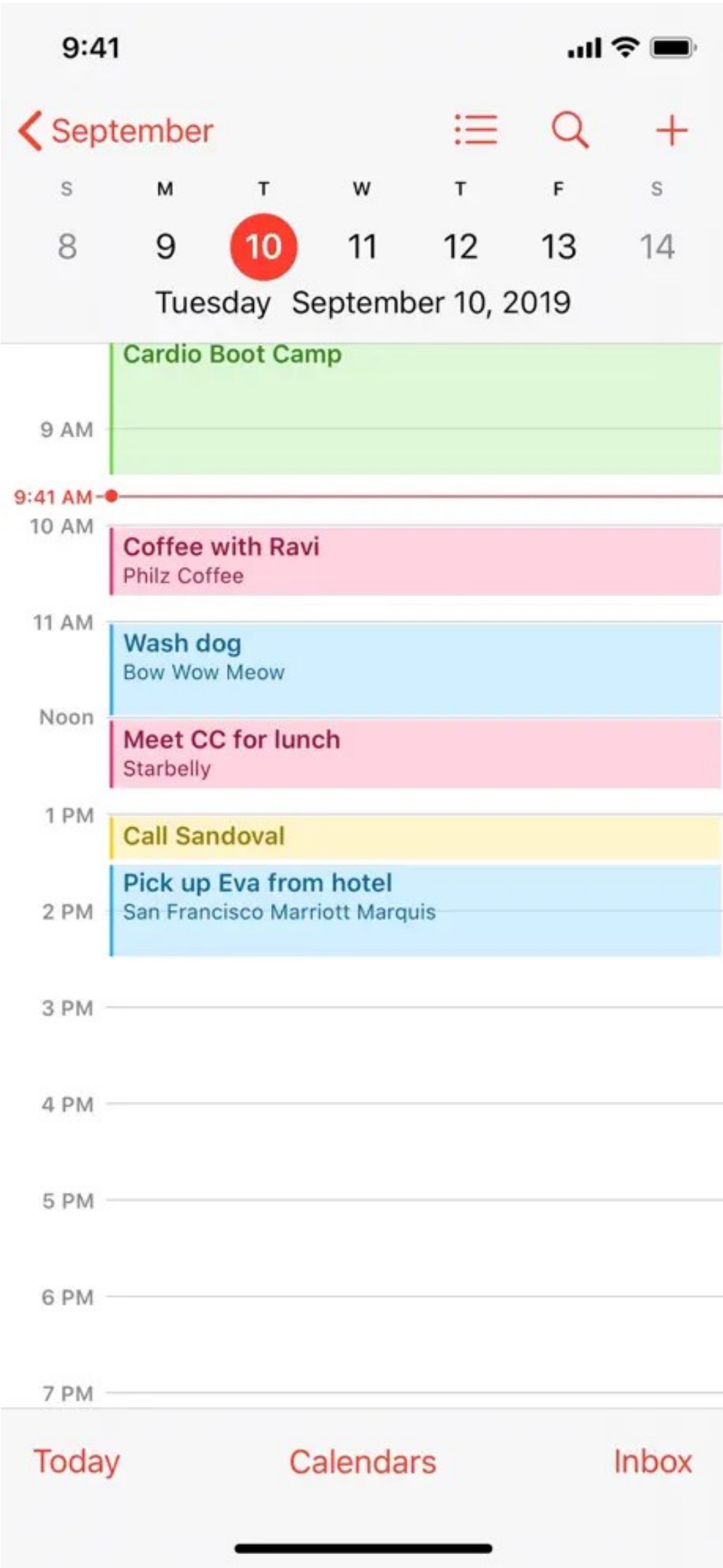
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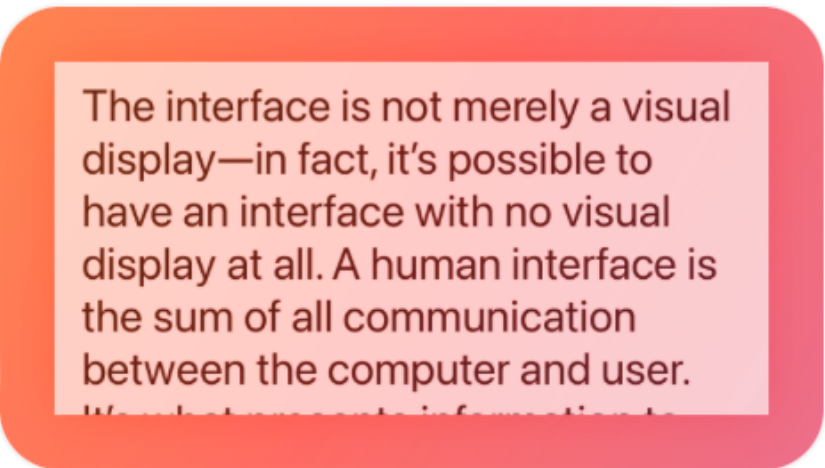
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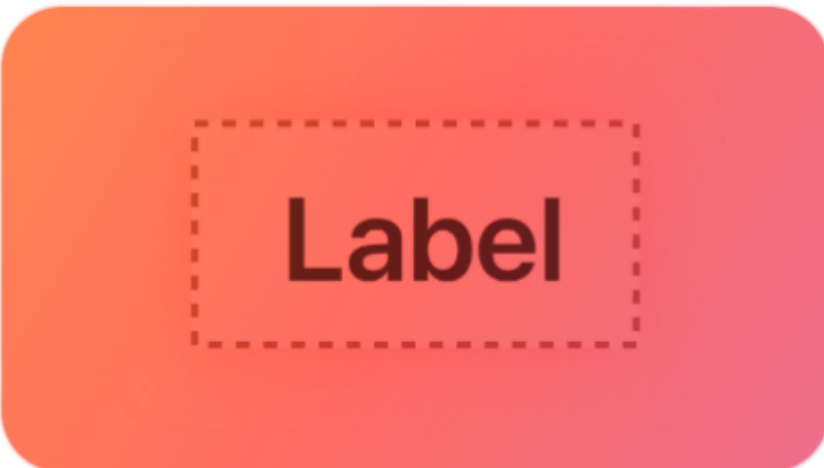
Calendar



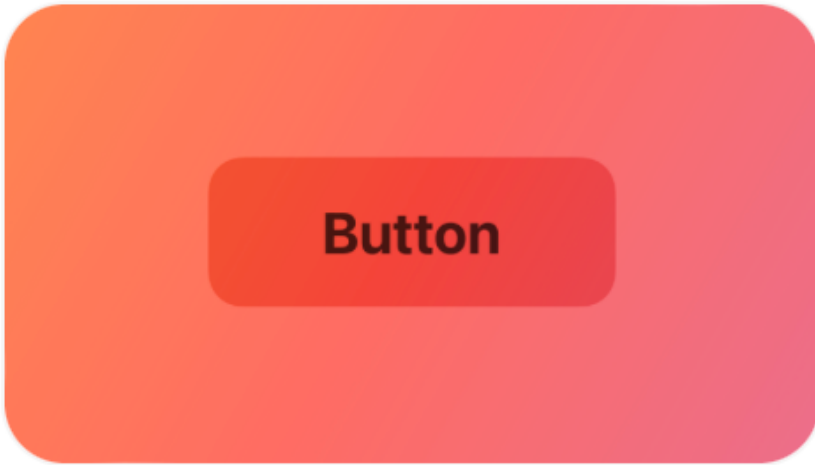
Navigation bars



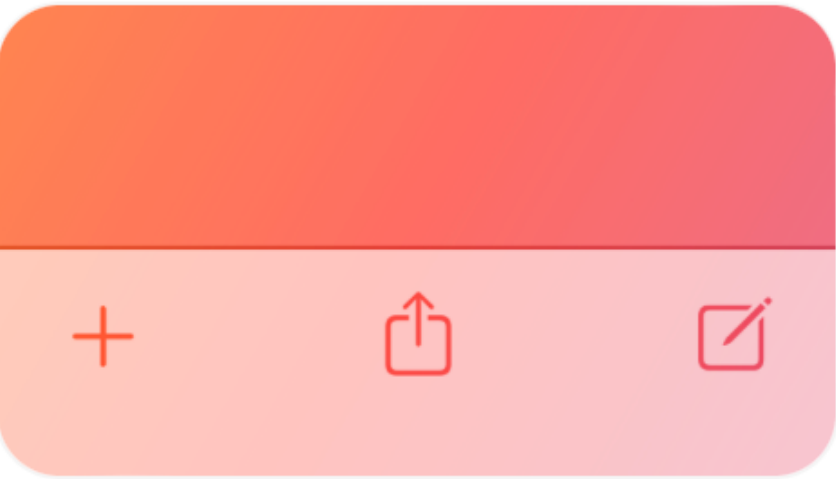
Text views



Labels



Buttons

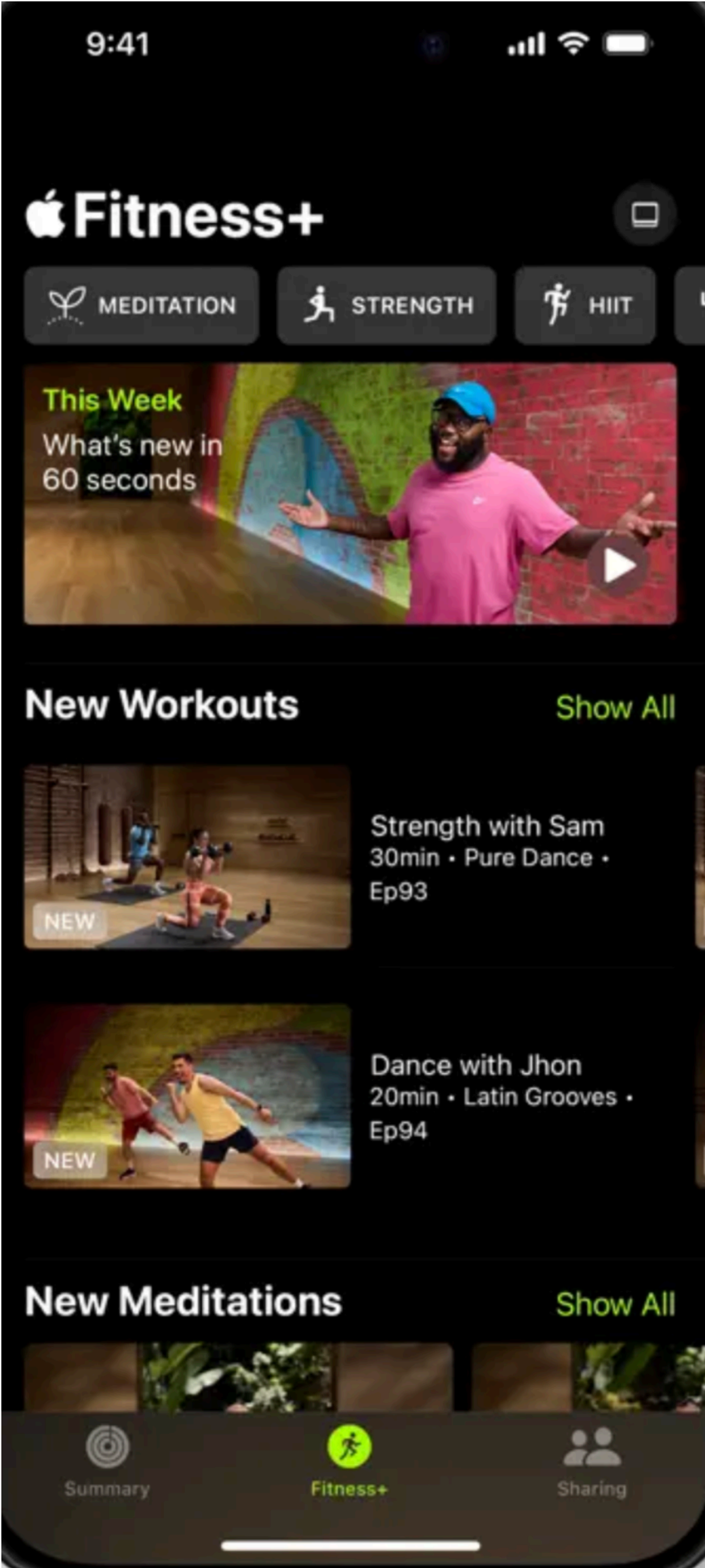


Toolbars

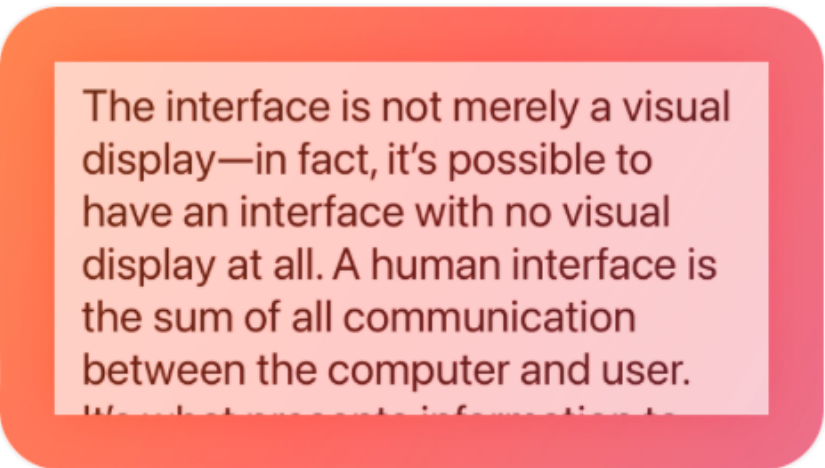


Boxes

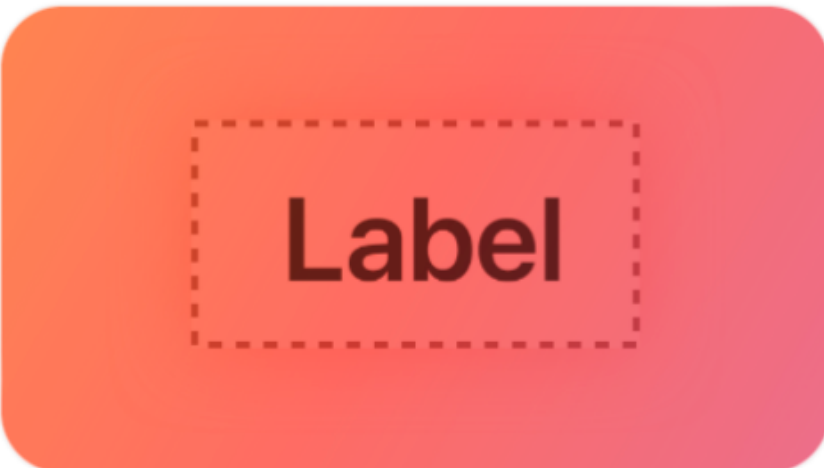
Fitness



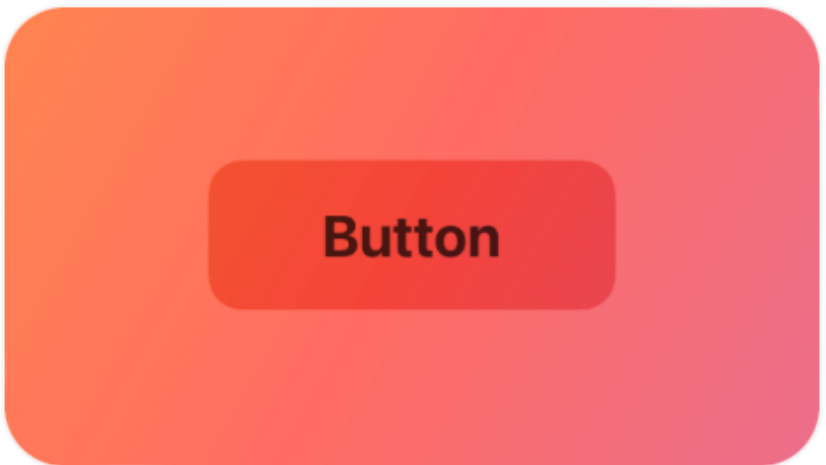
Navigation bars



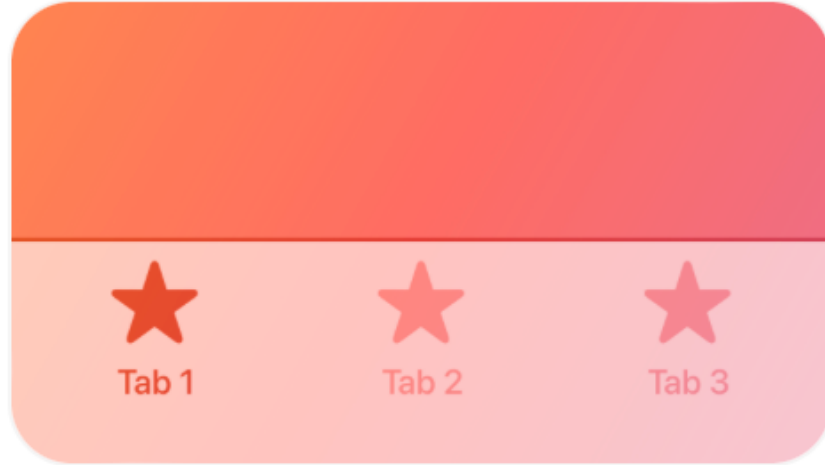
Text views



Labels



Buttons



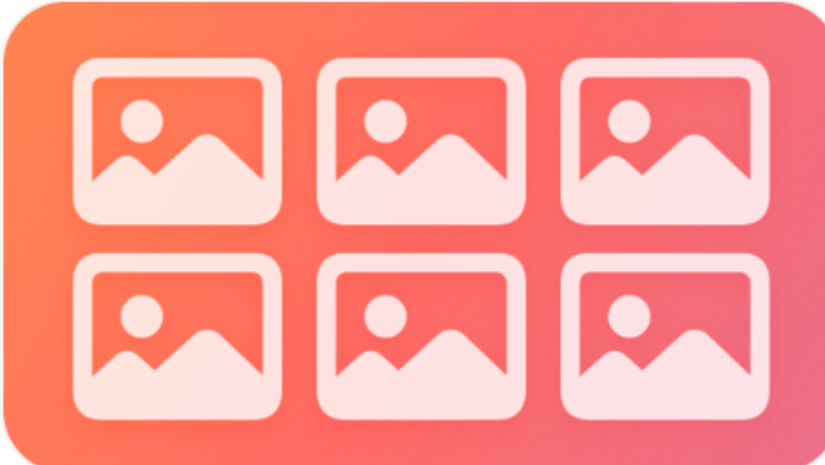
Tab bars



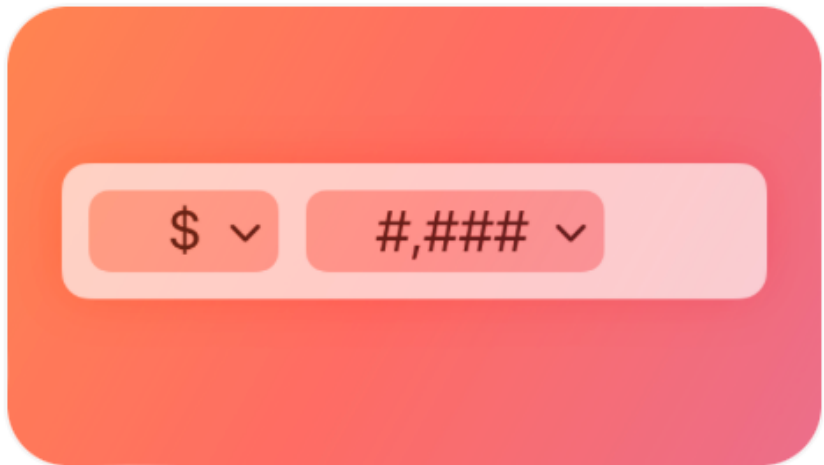
Boxes



Image views



Collections



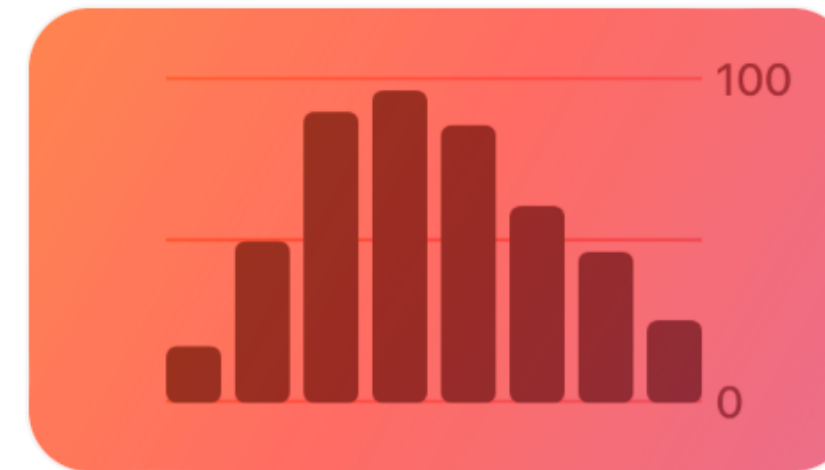
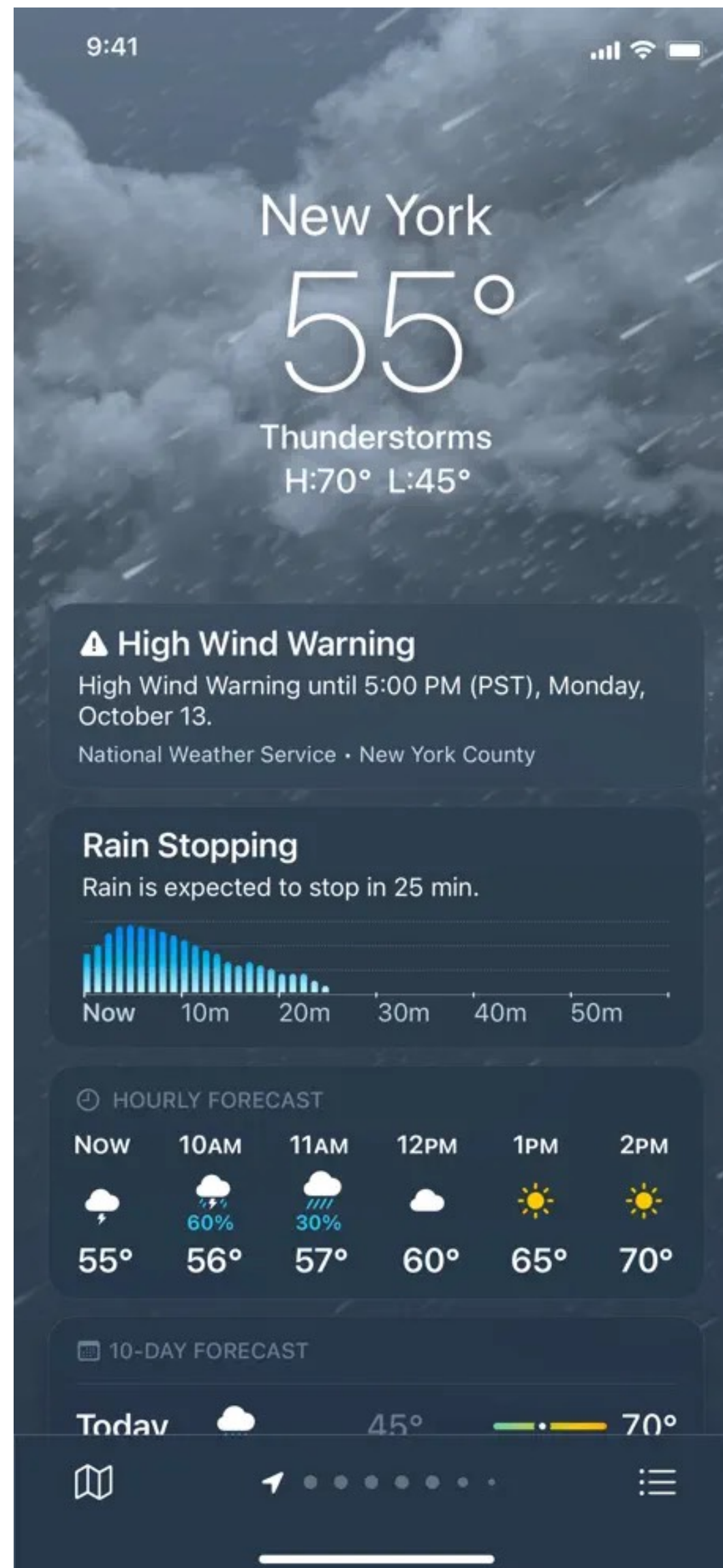
Token fields



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Weather



Charts

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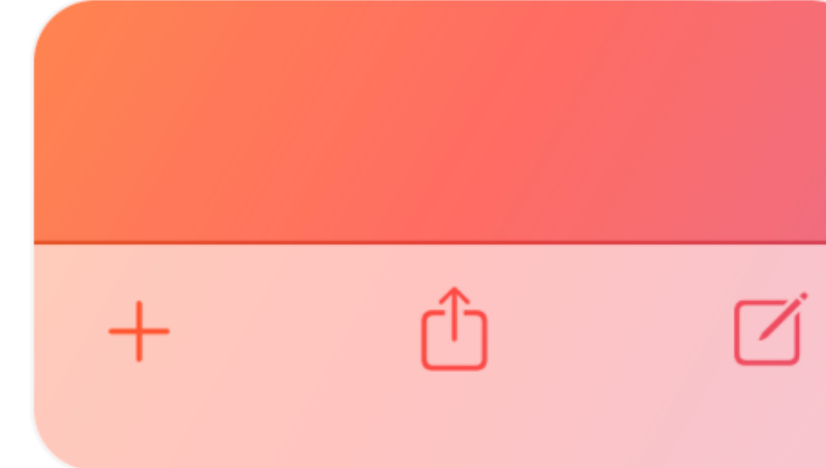
Text views

Label

Labels

Button

Buttons



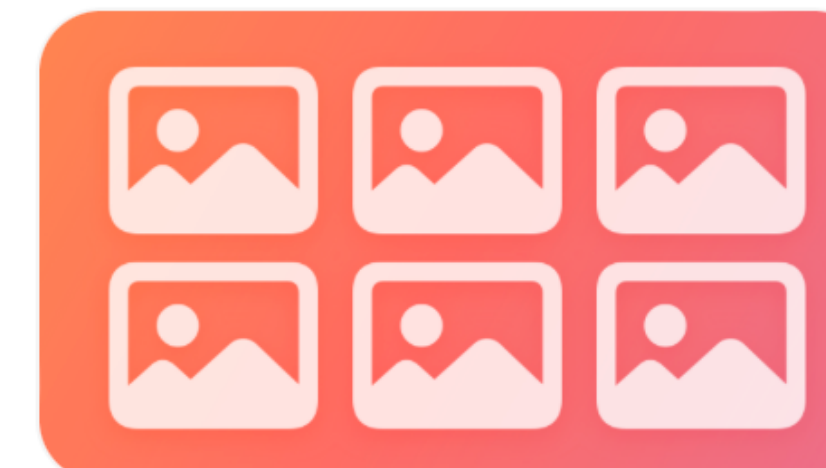
Toolbars



Boxes



Image views



Collections

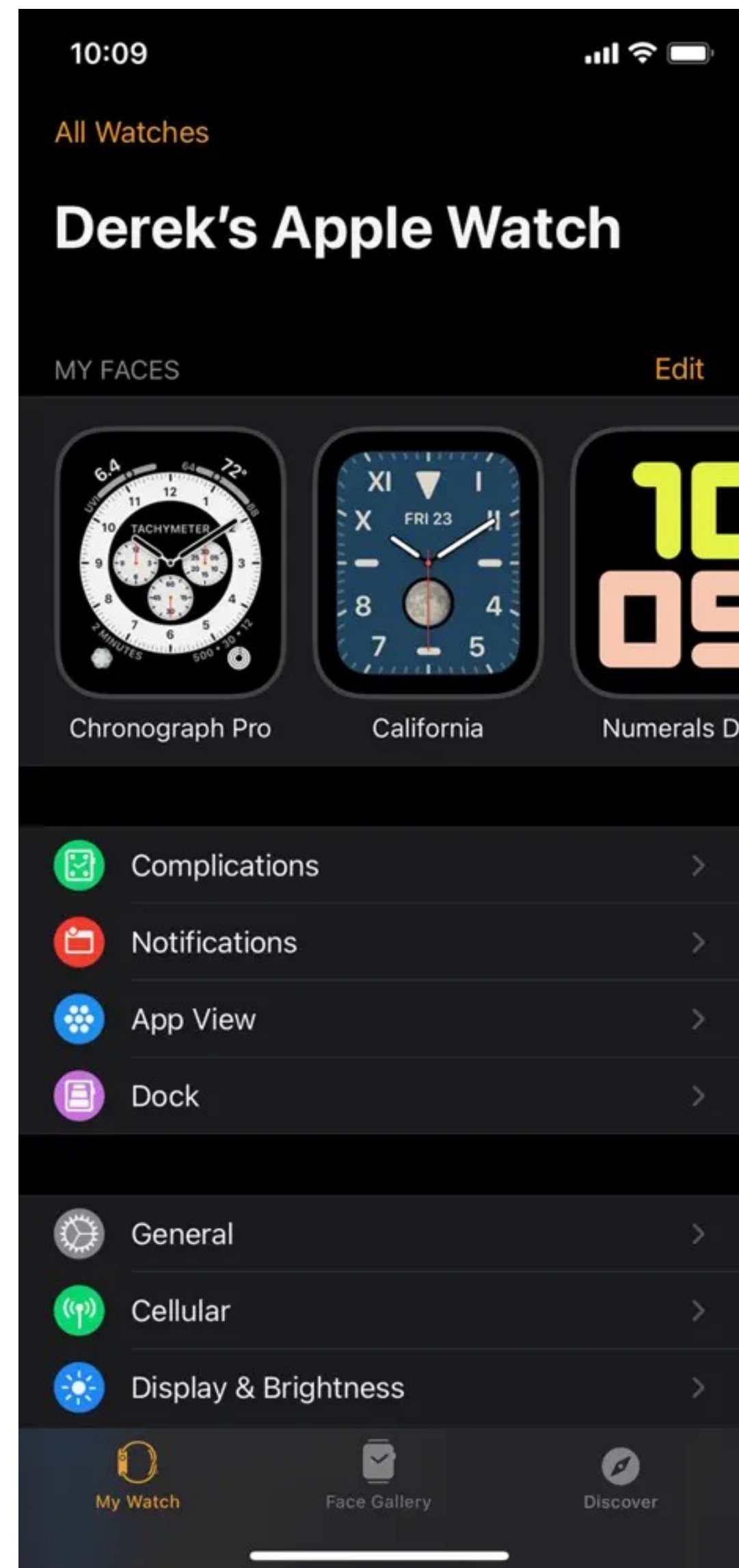


Page controls

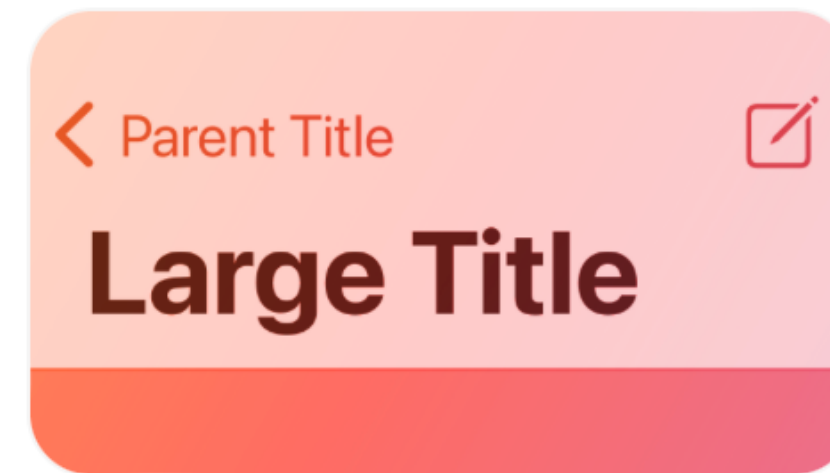


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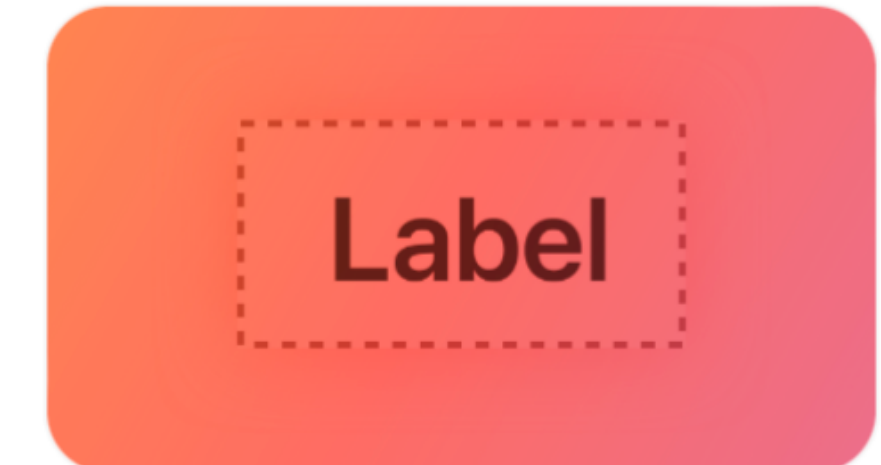
Watch



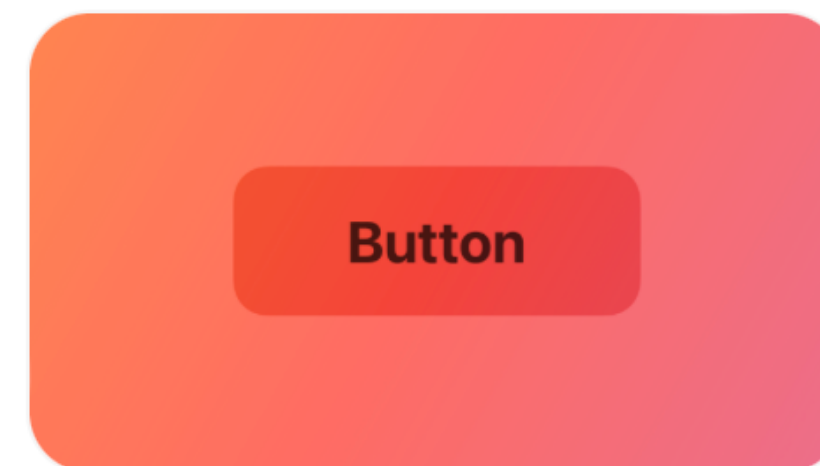
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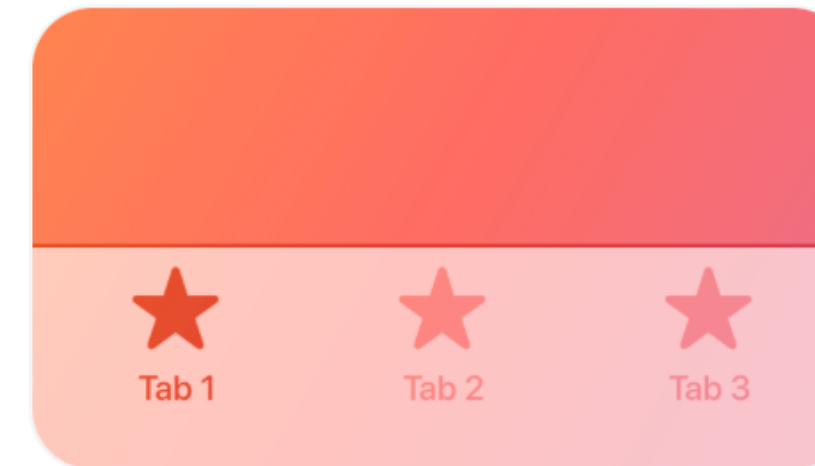
Image views



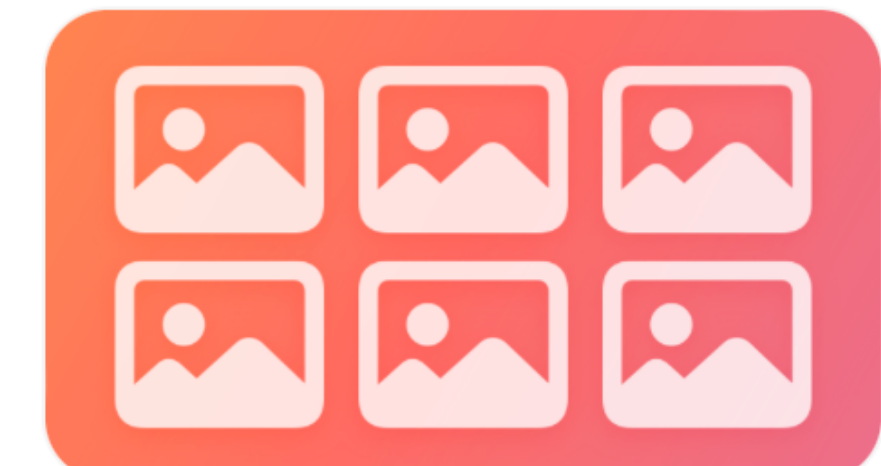
Labels



Buttons



Tab bars



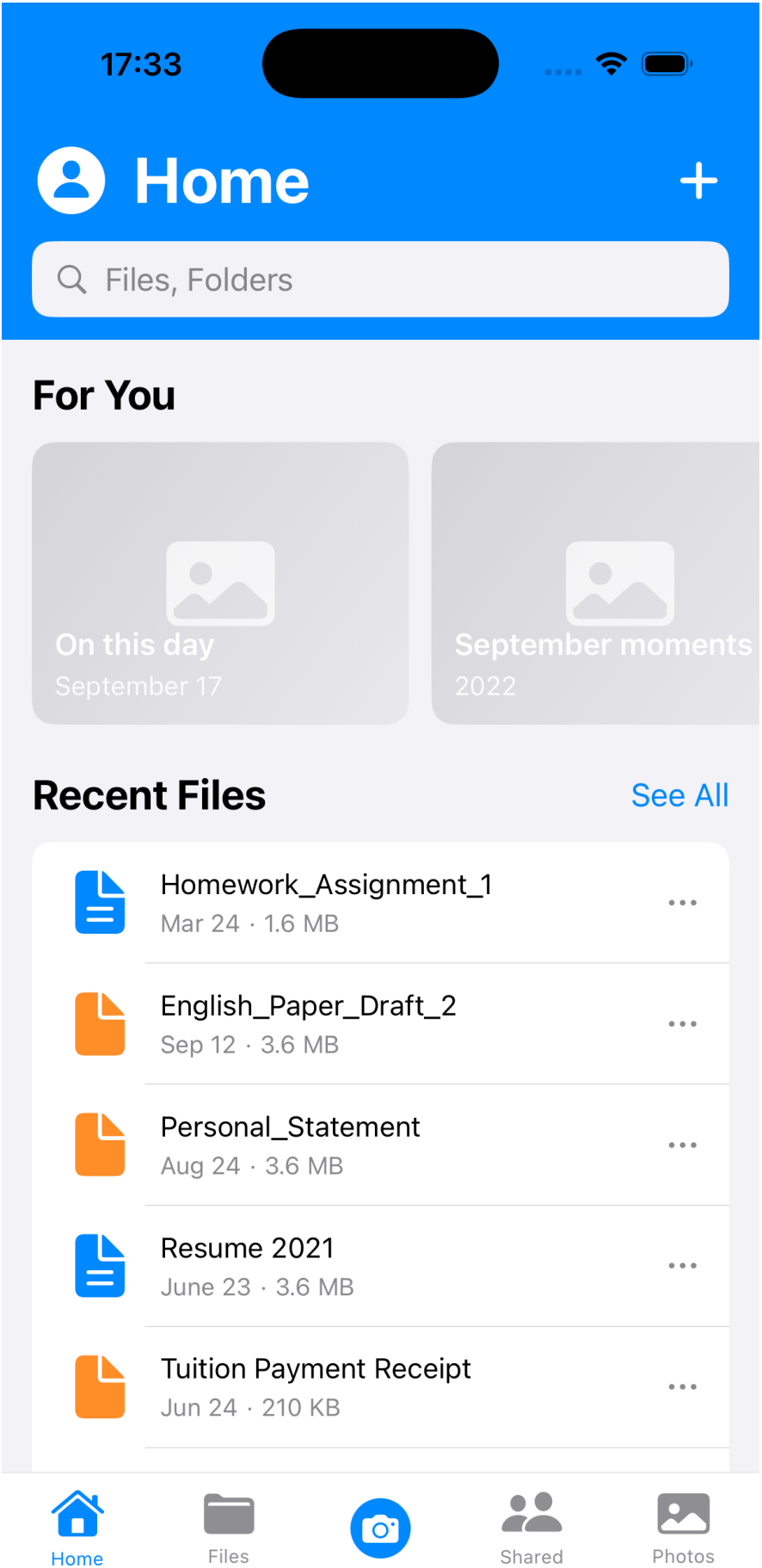
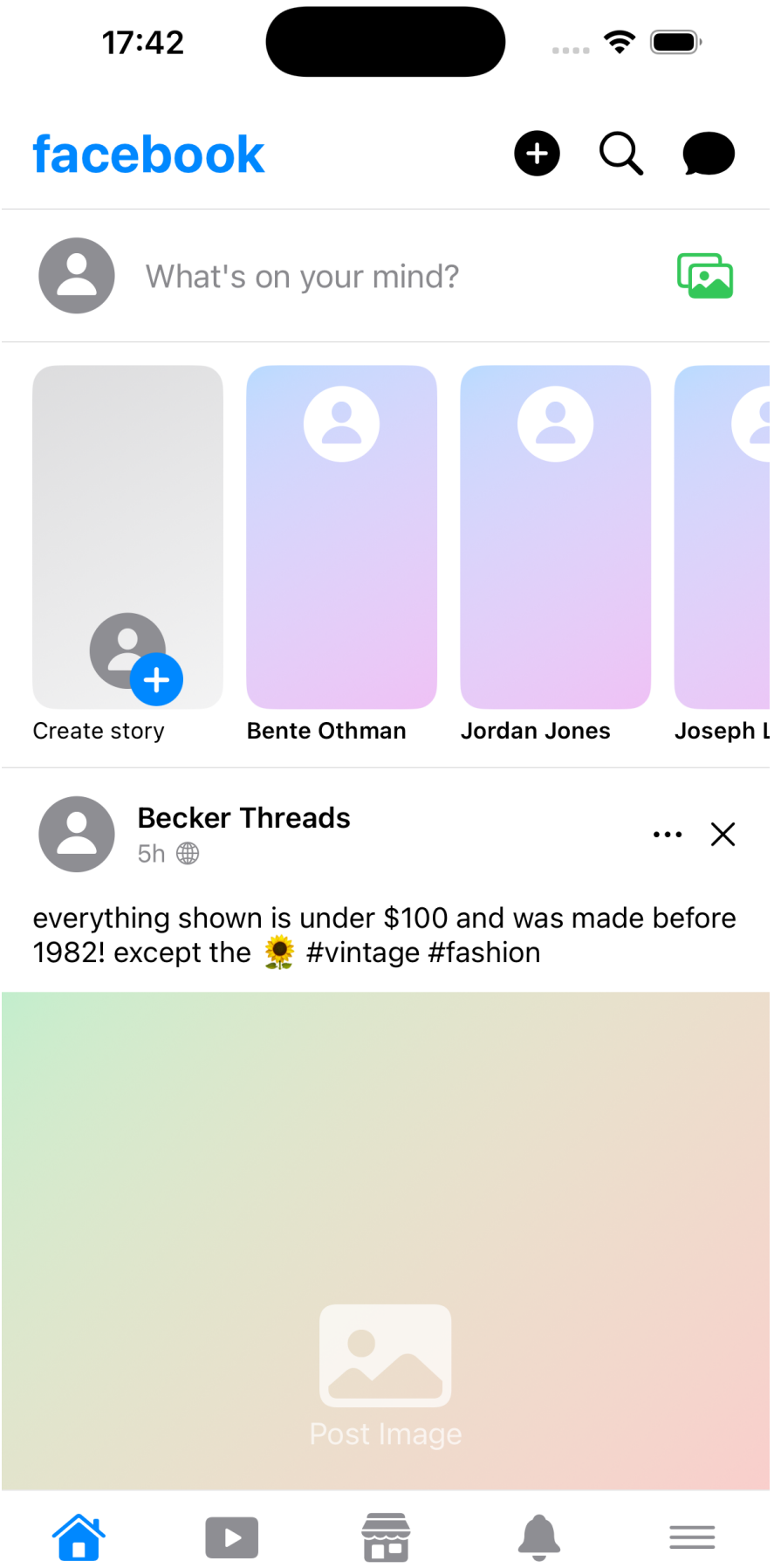
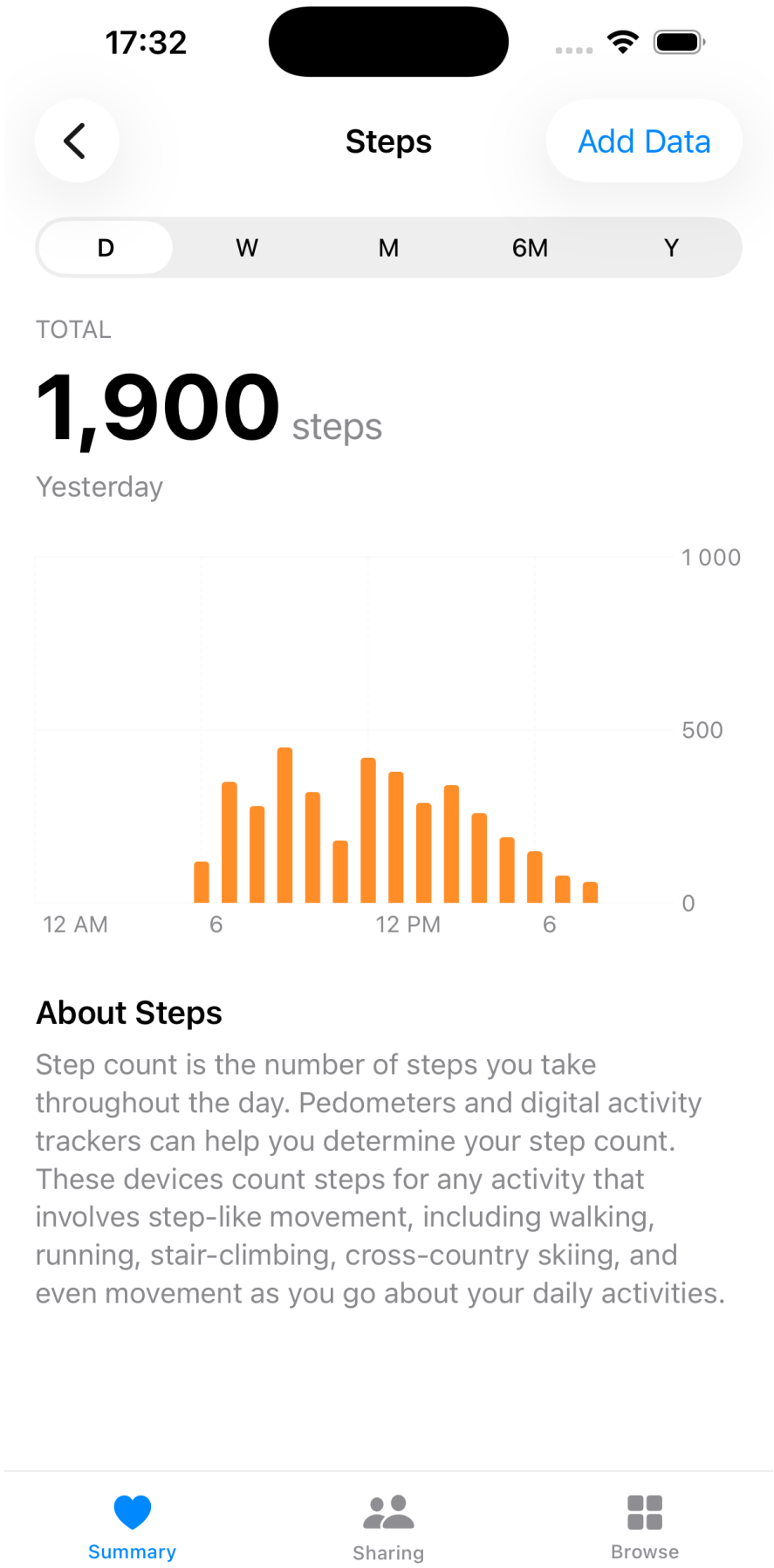
Collections



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DEMO





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Summary and Questions



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Useful links

1. **Apple Human Interface Guidelines:** <https://developer.apple.com/design/human-interface-guidelines>
 2. **Layout and organisations:** <https://developer.apple.com/design/human-interface-guidelines/layout-and-organization>
 3. **Selection and inputs:** <https://developer.apple.com/design/human-interface-guidelines/selection-and-input>
 4. **Navigation and search:** <https://developer.apple.com/design/human-interface-guidelines/navigation-and-search>
 5. **Menus and actions:** <https://developer.apple.com/design/human-interface-guidelines/menus-and-actions>
-



“What we know is a drop, what we
don't know is an ocean.”

Isaac Newton
